PREDICTING STUDENTS' DROPOUT AND ACADEMIC SUCCESS
ICT 110 - RxPulseMH
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DATASET DESCRIPTION

The project will use the Predict Students' Dropout and Academic Success dataset from the UCI Machine Learning Repository (Dataset ID: 697). The UCI repository lists 4,424 student records, 36 predictor features, and one target variable. Each record represents an individual student [1]. The dataset contains demographic, socioeconomic, admission, macroeconomic, and academic performance information from the first and second semesters. It was collected from the Polytechnic Institute of Portalegre in Portugal, covering academic years 2008/2009 to 2018/2019. The data came from several sources, including the Academic Management System (AMS), Support System for the Teaching Activity (PAE), Directorate-General for Higher Education (DGES), and Contemporary Portugal Database (PORDATA). It is publicly available under the Creative Commons Attribution 4.0 International (CC BY 4.0) license [1], [2].

Dataset Link:
https://archive.ics.uci.edu/dataset/697/predict%20students%20dropout%20and%20academic%20success

PROBLEM STATEMENT

Student dropout is a concern for higher education institutions because it affects academic progress and student completion rates. Identifying students who may eventually drop out can help academic administrators, advisers, and student support offices recognize academic patterns and provide appropriate assistance. This study aims to determine whether machine learning can classify students' eventual academic outcomes using their demographic, socioeconomic, admission, and academic performance information.

The project will use supervised multiclass classification because the target variable, Target, contains three predefined categories: Dropout, Enrolled, and Graduate. These labels represent each student's status at the end of the normal duration of their course. The model will use information available at enrollment and at the end of the first and second semesters. Therefore, predictions will be made after the second semester to estimate students' eventual academic outcomes rather than at initial admission

CHOSEN METRIC

The primary evaluation metric will be the Macro-Averaged F1-Score (Macro-F1). This is appropriate because the dataset has imbalanced classes: Graduate (49.93%), Dropout (32.12%), and Enrolled (17.95%) [2]. Macro-F1 gives equal importance to all three classes by averaging their individual F1-scores. This allows the model's ability to identify each academic outcome to be evaluated fairly. Per-class precision, recall, and a confusion matrix will also be examined.

BASELINE IDEA

The baseline will use the majority-class strategy, which predicts the most frequent category in the training data, expected to be Graduate. Based on the complete dataset, Graduate represents 2,209 of the 4,424 records, or approximately 49.93%. This baseline has an estimated Macro-F1 score of 0.222. Its actual performance will be calculated using the same cross-validation procedure and evaluation metric as the other machine learning models, with the majority class determined from each training fold only.

According to the original research paper, all data are anonymized, and compliance with the institution's Privacy and Personal Data Processing Policy is ensured in accordance with the General Data Protection Regulation (GDPR) [2]. However, demographic and socioeconomic factors may contribute to unequal model performance across student groups. Since the dataset comes from a single higher education institution and covers academic years 2008/2009 to 2018/2019, its findings may not fully represent students from other institutions or current academic populations. The model will be used solely for academic analysis and as supplementary decision-support information to help identify students who may need academic assistance, not as the sole basis for decisions or to restrict educational opportunities.

REFERENCES

[1] V. Realinho, M. Vieira Martins, J. Machado, and L. Baptista, "Predict Students' Dropout and Academic Success," UCI Machine Learning Repository, 2021, doi: 10.24432/C5MC89.

[2] V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting Student Dropout and Academic Success," Data, vol. 7, no. 11, Art. no. 146, 2022, doi: 10.3390/data7110146.